



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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FACULTY OF COMPUTING, UTM
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PROJECT 1

SECI 1143 : PROBABILITY AND STATISTICAL DATA ANALYSIS

SECTION 01

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COURSE : BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

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1.0 INTRODUCTION

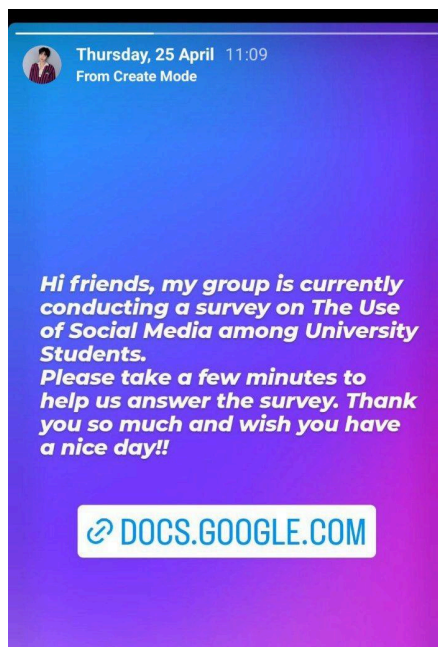
Nowadays, social media has become an important part of our daily life, it profoundly impacts various aspects of society, including communication, education and entertainment. Understanding the importance of social media among university students was the goal of this study.

To determine the responses to the research question and analyze the outcomes, the study gathers data using the questionnaire approach. The title of this report is Survey on The Use of Social Media among University Students. The general aim of this study is to investigate and observe how social media impacts students' daily life. This survey is mainly focusing on all university students in Malaysia.

At the end of our survey, we expect that university students use social media for various purposes in their daily life. In addition, we anticipate that university students will be highly self-disciplined and able to use social media primarily for positive and constructive endeavors.

Methodology

A questionnaire is used in the research since it fulfills the prerequisites mentioned above and makes it possible to quickly and efficiently collect a significant number of data. In order to ensure that the replies collected are from the appropriate respondent that from all universities in Malaysia, it was decided to launch this survey online using a Google Form and to distribute it through social media platforms like Telegram, Whatsapp, Instagram, and others. This survey was completed in a week by a total of 61 participants.



Survey link:

https://docs.google.com/forms/d/e/1FAIpQLScZ29xsb5jP9_YoaE-7FkpSwTsnSfYhTLNZmWybsHdwvknNMw/viewform?usp=pp_url

Questionnaire

SEC1143-01 Probability & Statistical Data Analysis (PSDA)

Hello everyone! 🙋 We are UTM students from SEC1143 Probability & Statistical Data Analysis Section 1 and now we are conducting an online survey which aims to investigate the use of social media among university students. This study aims to understand the patterns, preferences, and impacts of social media usage on students' academic and personal lives.

We kindly request a few minutes of your time to fill out the questionnaire at your convenience. This questionnaire consists of 2 parts. Please ensure all questions are fully completed. We prioritize your privacy and security all data provided will be kept confidential and secure.

Your participation is greatly appreciated and will contribute to the success of our project. Thank you for your cooperation! 🙏❤️

tanzhaohong@graduate.utm.my [Switch account](#)



* Indicates required question

Email *

☒ Record tanzhaohong@graduate.utm.my as the email to be included with my response

Part 1

Gender *

- ☐ Male
☐ Female

Age *

Your answer

Race *

- ☐ Malays
☐ Chinese
☐ Indians
☐ Other: _____

University (Example: UTM) *

Your answer

Latest CGPA (Example: 4.00) *

Your answer

[Back](#)

[Next](#)

[Clear form](#)

Part 2

How often do you spend your time daily for the following purposes? *

	Never	Rarely	Sometime	Often	Always
Education	☐	☐	☐	☐	☐
Communication	☐	☐	☐	☐	☐
Entertainment	☐	☐	☐	☐	☐
Online shopping	☐	☐	☐	☐	☐
Information and News	☐	☐	☐	☐	☐

How many social media application do you have? (Example: 10) *

Your answer

Amount of time spent on social media platforms per day (Example: 3 hours) *

Your answer

Which social media application do you used the most? *

- ☐ Facebook
☐ Instagram
☐ Twitter
☐ Tik Tok
☐ Snapchat
☐ YouTube
☐ Telegram
☐ WhatsAoo

How many content do you post on social media per day? (Example: 3) *

Your answer

What is the highest amount of friends or followers among your social media platforms *

Your answer

How would do you rank the importance of social media in you daily life? *

Least important 1 2 3 4 5 Most important
☐ ☐ ☐ ☐ ☐

How often do you share the contents on social media? (text post, image, video, etc) *

Never 1 2 3 4 5 Always
☐ ☐ ☐ ☐ ☐

[Back](#)

[Submit](#)

[Clear form](#)

2.0 DATA COLLECTION

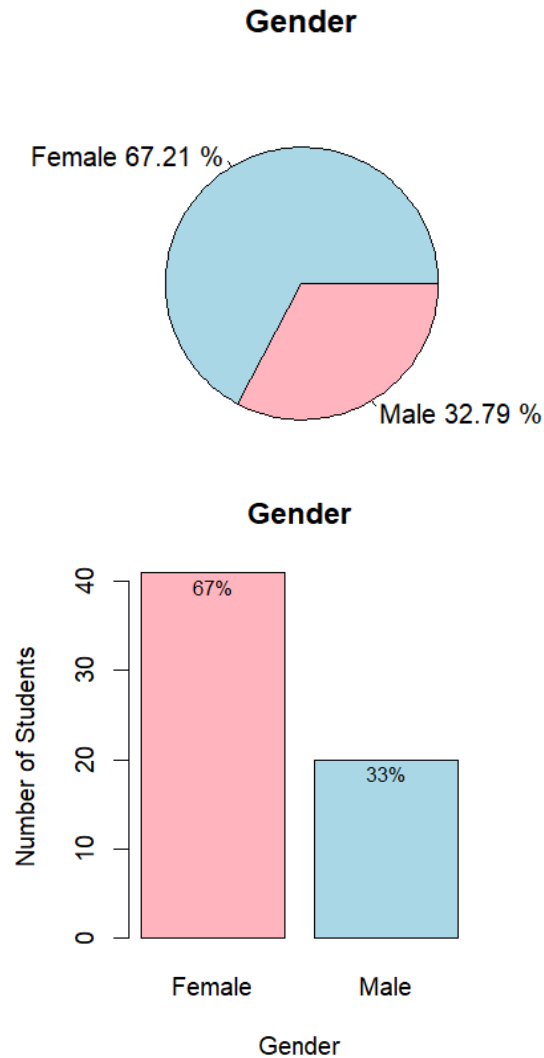
Data is collected via a questionnaire that is distributed using a Google Form on various social media platforms. The information is successfully collected from 61 respondents in total.

The questionnaire is divided into two parts: personal information and use of social media. Each element contains a set of related variables that must be obtained from the respondents. The variables that gathered in each part are shown in the table below:

Variable	Level of Measurement
Personal Information 1. Gender 2. Age 3. Race 4. University 5. Latest CGPA	Nominal Ratio Nominal Nominal Interval
Use of Social Media 1. How many social media apps do you have? 2. Amount of time spent on social media platforms daily. 3. What social media app do you usually use the most? 4. How many content that you post on social media per day? 5. What is the highest amount of friends or followers among your social media platforms? 6. How would you rank the importance of social media in your daily life? 7. How often do you share the contents on social media? (text post, image, video) 8. Purpose of using social media.	Ratio Interval Nominal Ratio Ratio Ordinal Ordinal Ordinal

3.0 DATA ANALYSIS

3.1 Gender

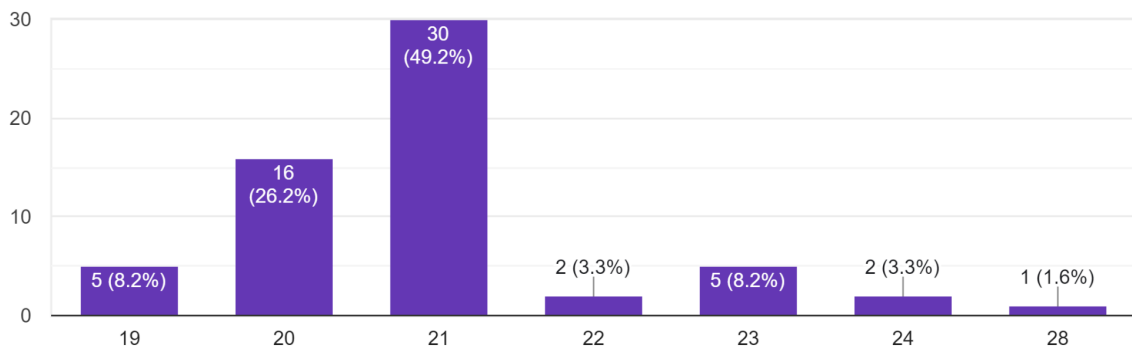


Based on the pie chart and bar chart above, we had collected 61 responses. Among the 61 respondents, there are 41 females and 20 males. The percentages of females and males are 67.2% and 32.8% respectively.

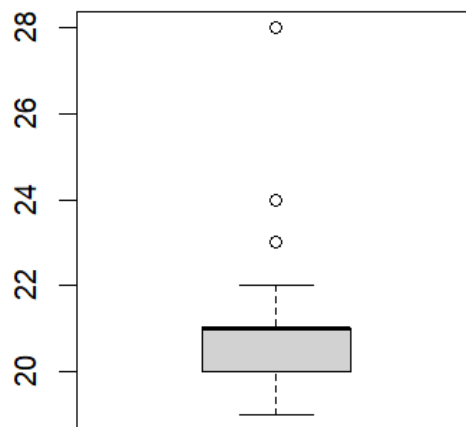
3.2 Age

Age

61 responses



Boxplot of Age)



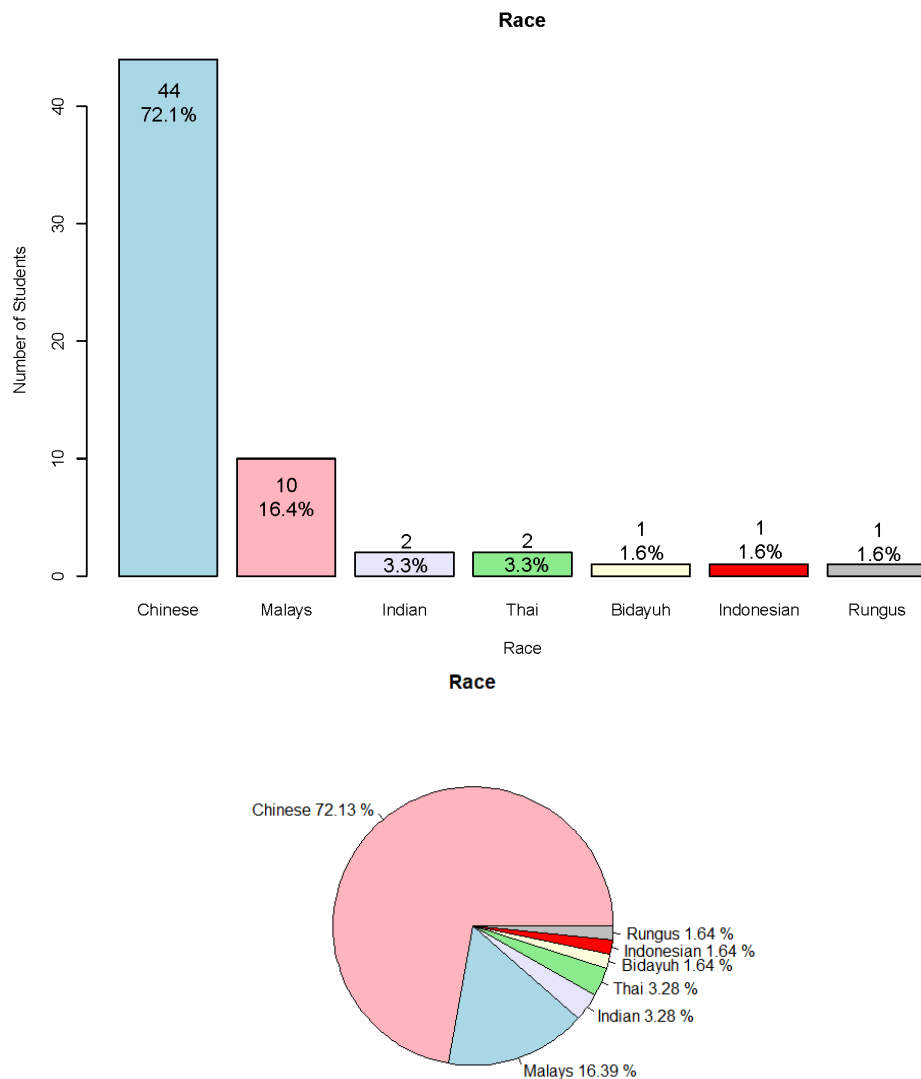
Out of 61 responses collected, the highest frequency is observed at age 21 with 30 occurrences, which accounts for 49.2% of total responses. Following closely, age 20 also shows a relatively high frequency, with 16 occurrences making up 26.2% of the responses. Conversely, certain age groups such as age 19 and age 23 exhibit lower frequencies, with the same occurrences which is 5, constituting 8.2% of total responses. Ages 22, 24 and 28 have the lowest frequencies or are even absent from the dataset. There are 2 respondents at ages 22 and 24 each with percentage 3.3% while there is only one respondent with an age of 28 with percentage 1.6%.

Based on the boxplot, it shows that it is positively skewed. The mean of age is 20.98, while the mode is age 21.

Quartile:

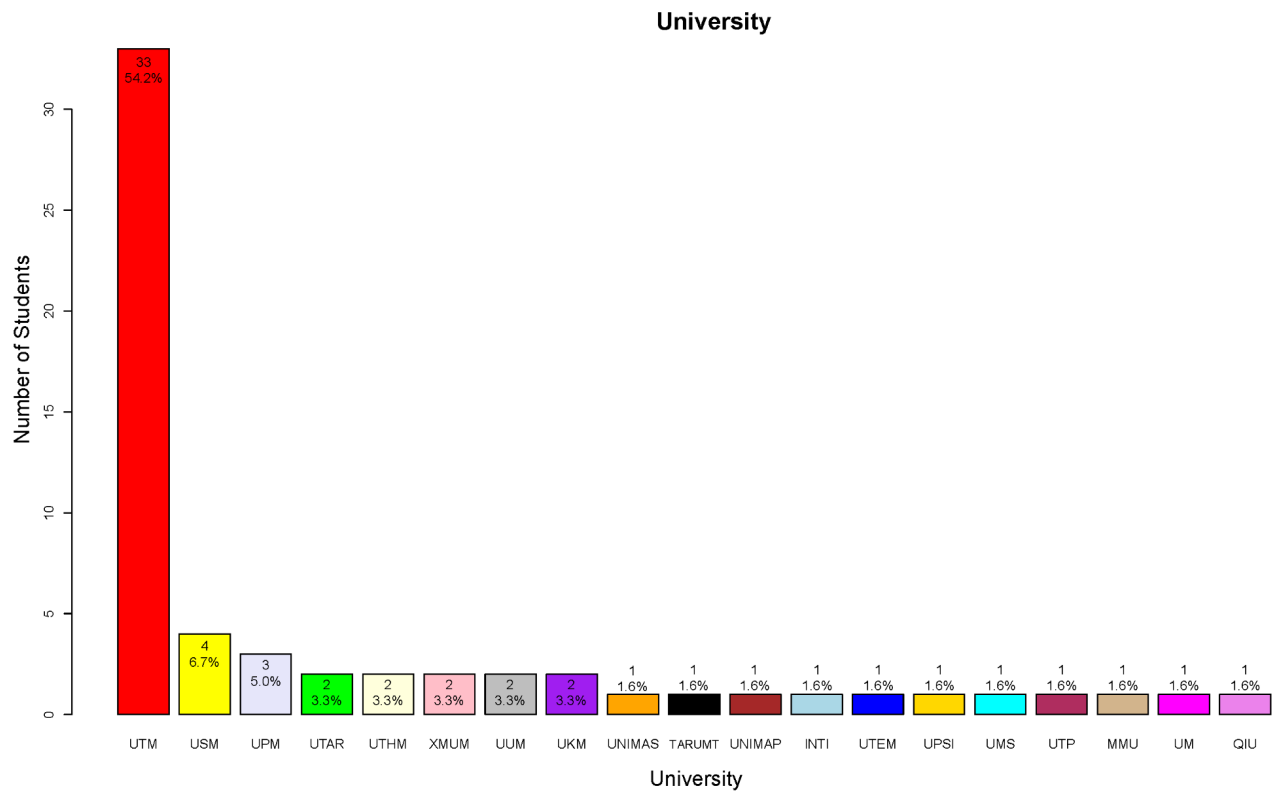
0%	25%	50%	75%	100%
19	20	21	21	28

3.3 Race



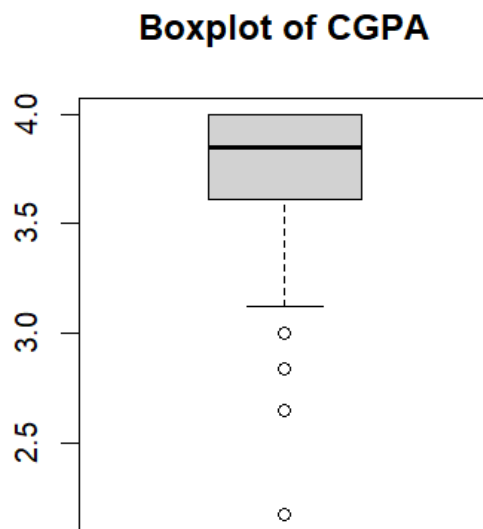
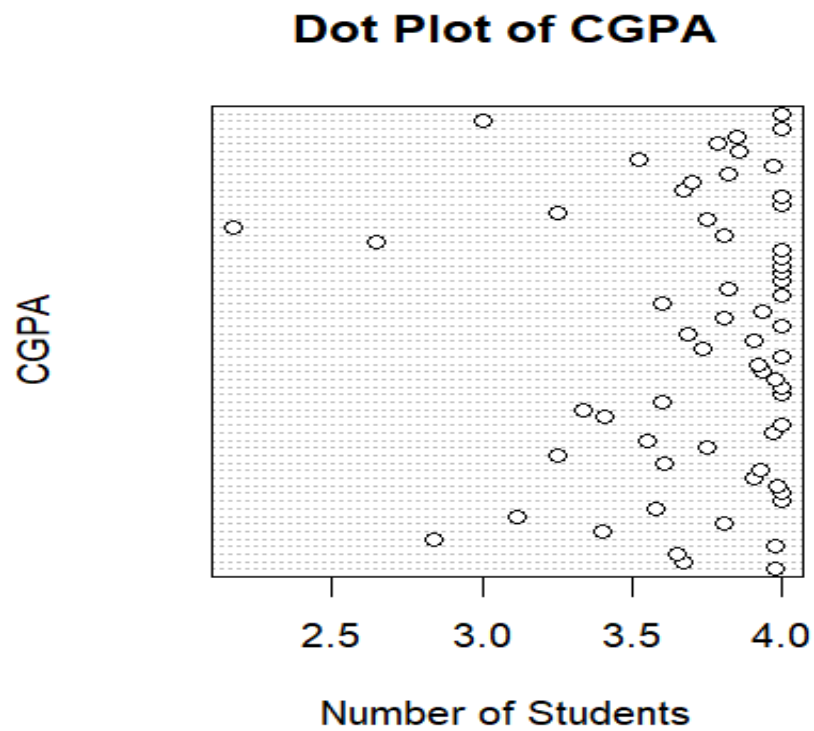
Out of 61 responses, the Chinese respondents make up the majority, totaling 44 with 72.1%. The second largest group is Malay, comprising 10 out of 61 responses with 16.4%. Following closely are Indian and Thai respondents, each with 2 responses, making up 3.3% of the total respondents. There is also one respondent from Sarawak, belonging to the Bidayuh ethnic group, accounting for 1.6%. Additionally, there are respondents from other ethnicities, including Rungus and Indonesian, each with 1 occurrence, constituting 1.6% of total responses..

3.4 University



As our target respondents are university students from Malaysia, there is a notable presence of various universities in our dataset. The majority of respondents hail from Universiti Teknologi Malaysia (UTM), totaling 33 participants, which accounts for 54.2% of the total. This prevalence can be attributed to our affiliation with UTM and our focused distribution of the questionnaire among UTM students. Following UTM, the second largest group of participants comes from Universiti Sains Malaysia (USM), with 4 respondents, constituting 6.7% of the total, followed by Universiti Putra Malaysia (UPM) with 3 respondents, making up 5.0% of the total responses. Additionally, there is representation from several other universities, including UTAR, UTHM, XMUM, UUM, and UKM, each contributing 2 responses, accounting for 3.3% each. Furthermore, there is one participant from each of the following universities: UNIMAS, TARUMT, UNIMAP, INTI, UTEM, UPSI, UMS, UTP, MMU, UM, and QIU, each representing 1.6% of the total responses. In summary, data was collected from 12 local universities and 7 private universities.

3.5 CGPA



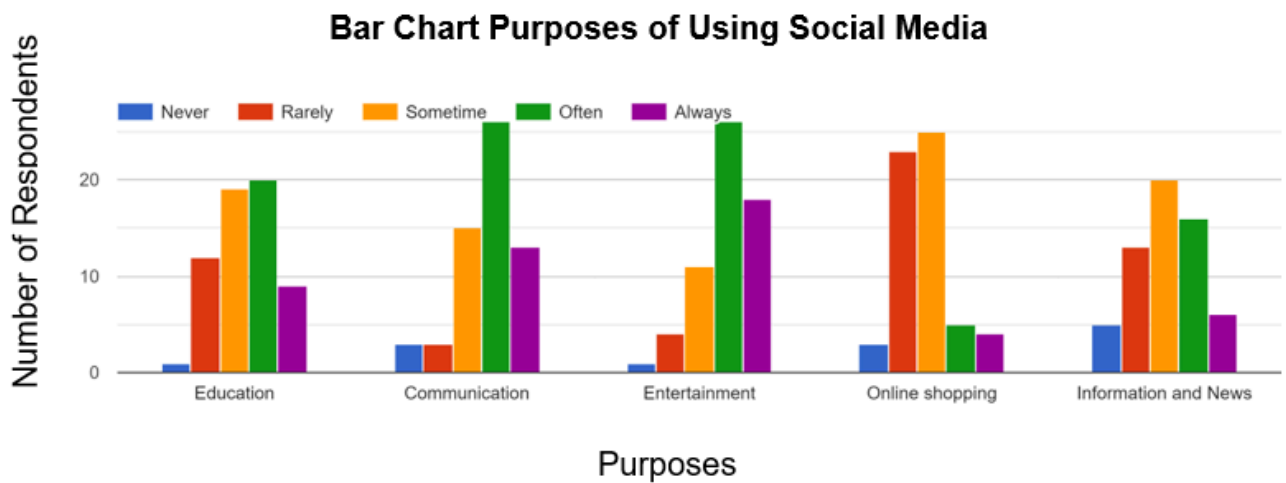
Based on the dot plot above, it shows that most of the respondents have a CGPA above 3.50. Only a few respondents have CGPA below 3.00.

Based on the boxplot, it is skewed to the left. From all the CGPA, it has a mean of 3.73 and a mode of 4.00.

Quartile:

0%	25%	50%	75%	100%
2.17	3.61	3.85	4.00	4.00

3.6 Purposes of Using Social Media

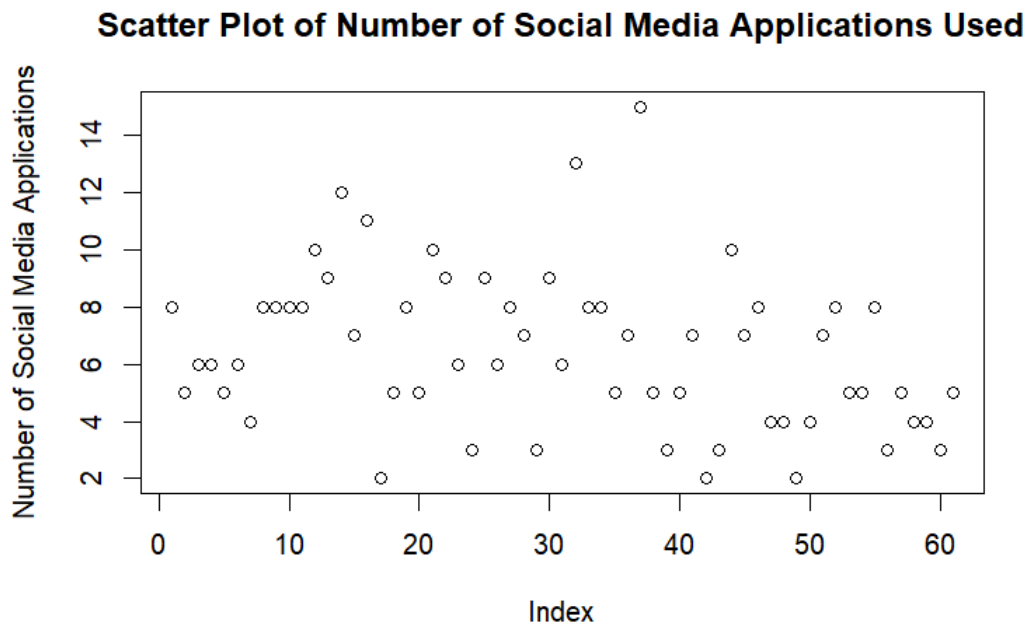


From the bar chart above, it shows how often respondents spend their time daily for the following 5 purposes which are education, communication, entertainment, online shopping and information and news.

For education purposes, it shows that 2% which is 1 out of 61 respondents never use social media for education purposes, 33% which is 20 out of 61 respondents often use it for education purposes. Besides that, there are 5% which is 3 out of 61 respondents who never and rarely use social media for communication purposes whereas there is 43% which is 26 out of 61 respondents who often use it for communication. Moreover, 2% of respondents never use it for entertainment and 43%, 26 respondents said that they often use social media applications for entertainment. Furthermore, for online shopping purposes, 5% of respondents never use social media for their online shopping and 41% which is 25 out of 61 respondents said that they sometimes use it only. Lastly, there is 8% which is 5 from those respondents who never get information and news through social media whereas there is 33% which is 20 from those 61 respondents who sometimes will get information and news through social media platforms.

To sum up from the graph above, mostly respondents used social media for their daily entertainment. Then followed by communication, education, information and news purposes. Most respondents less used social media for their online shopping. This might be because online shopping has their own platforms for only shopping.

3.7 Number of Social Media Applications Used



From the scatter plot above, it shows that all respondents have at least 2 social media applications accounts, and the most number of social media applications can be up to 15 from the responses of this survey.

Besides that scatter plot, box plots that show on the right side which are done by using RStudio code also show the mean value, mode value, and quartiles.

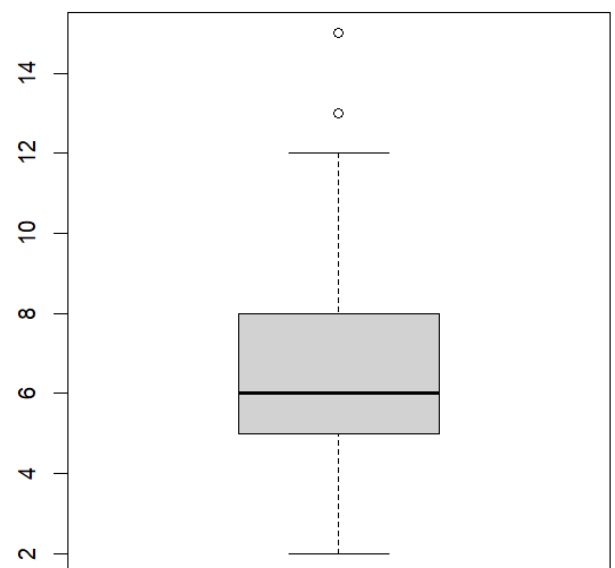
Mean value : 6.45901639344262

Mode value : 8

Quartiles :

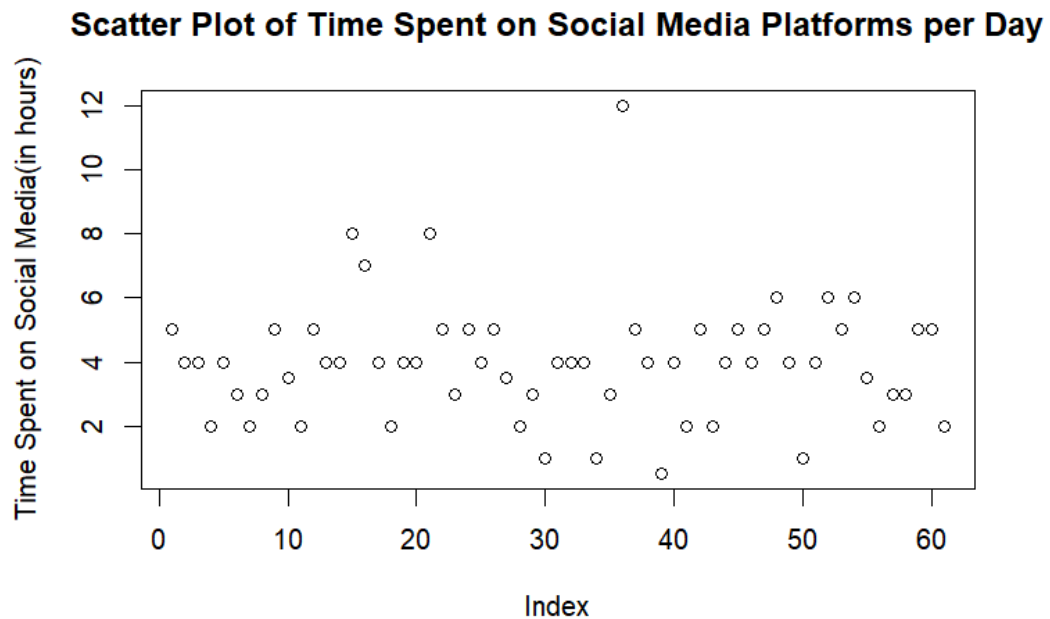
0%	25%	50%	75%	100%
2	5	6	8	15

Number of Social Media Applications Used



From the above graphs, it shows many people have 8 social media applications and average people will have 6 social media applications on their device for education, communication, entertainment, online shopping or information and news purposes.

3.8 Time Spent on Social Media Platforms per Day



Based on the scatter plot above, it shows that one of the respondents uses 12 hours on social media platforms per day. Most of the respondents use 4 to 5 hours per day on social media.

Besides that, box plots that show on the right side which are done by using RStudio code have shown the mean value, mode value, and quartiles.

Mean value : 3.98360655737705

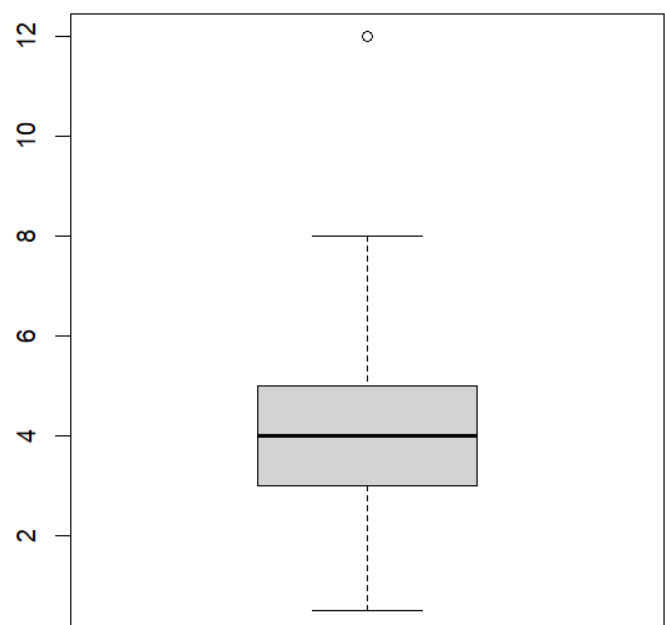
Mode value : 4

Quartiles :

0%	25%	50%	75%	100%
0.5	3.0	4.0	5.0	12.0

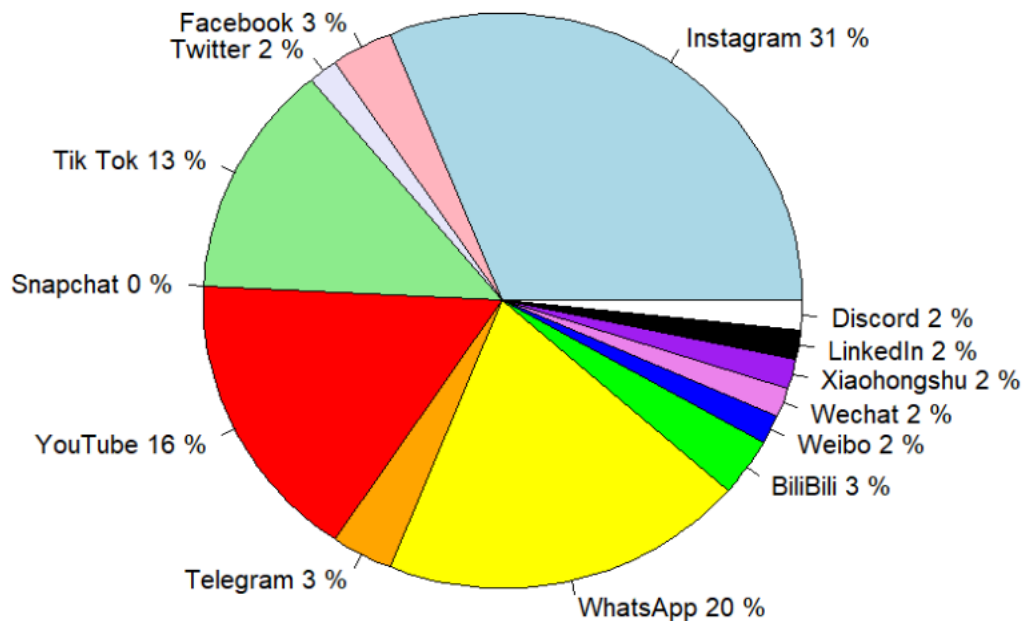
From the above graphs, it shows many people will spend 4 hours using social media.

Boxplot of Time Spent On Social Media (in Hours)



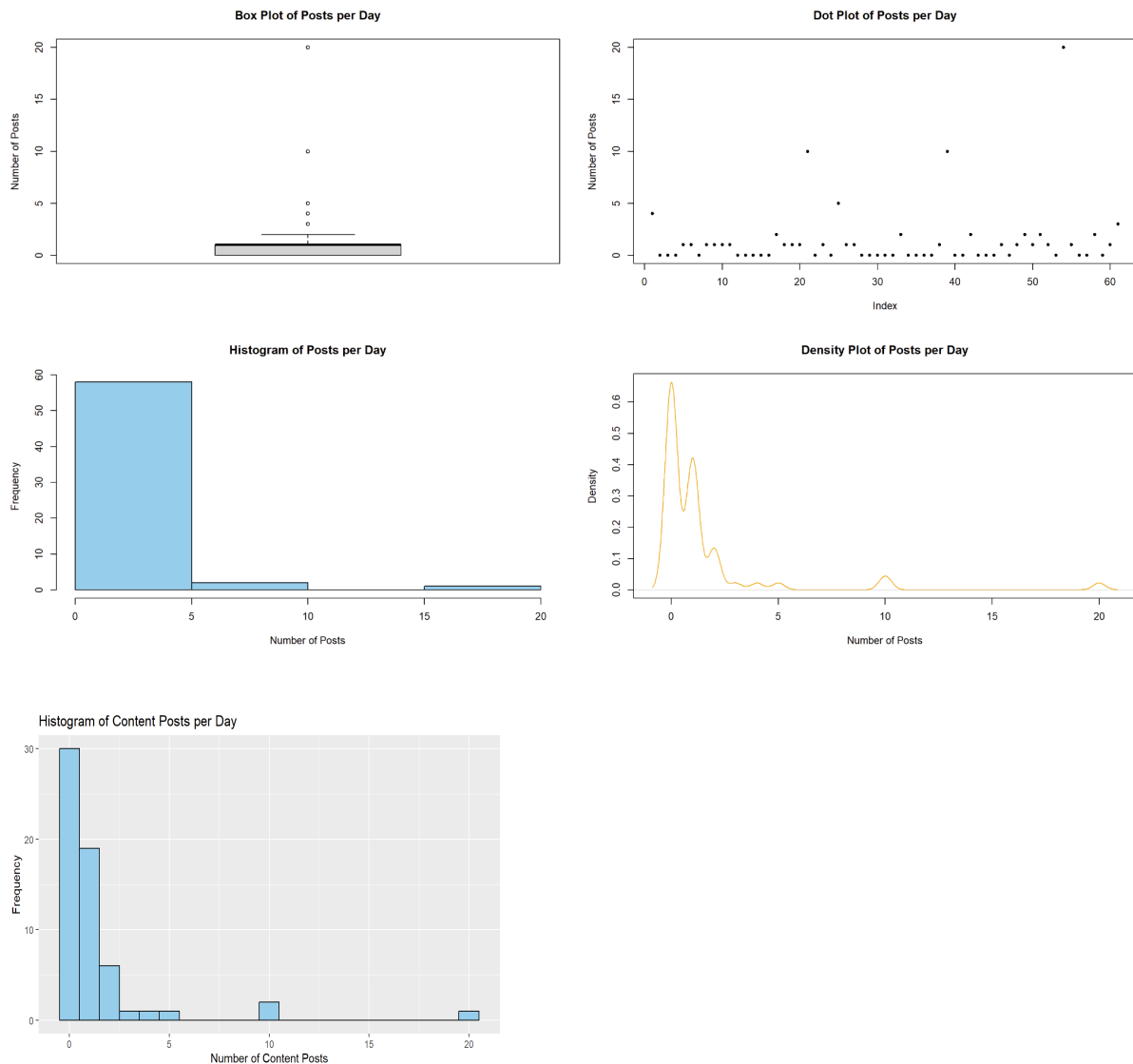
3.9 Social Media Platform That Used Most Frequently

Social Media Platform Used Most Frequently (Pie Chart)



Based on this pie chart, it clearly shows that the social media platform that the respondents used most frequently is Instagram(31%) which has 19 out of 61 respondents . Whereas Snapchat (0%) is the least frequently used by those respondents, because it shows no respondent response on it. The second most used social media platform is WhatsApp, 20% respondents choose it as the most used social media platform. Following Youtube which has 16%, then Tik Tok has 13% response. There are 3 social media platforms that get the same response which is 3% response which are Twitter, Telegram and BiliBili. Lastly, there are 2% responses that they use most frequently on different 5 social media platforms which are Discord, LinkedIn, Xiaohongshu, Wechat and Weibo.

3.10 Content that you post on social media per day



```
summary(posts_per_day)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	1.000	1.361	1.000	20.000

- The minimum number of content posted per day is 0.
- The 25th percentile (first quartile) is also 0.
- The median is 1.
- The mean is approximately 1.361.
- The 75th percentile (third quartile) is 1.
- The maximum number of content posted per day is 20.

The majority of days have either no content posted or very few posts. The average number of posts per day is slightly above 1. There's a wide range of variability, as indicated by the maximum value of 20 posts per day.

3.11 What is the highest number of friends or followers among your social media platforms?

> summary(data)

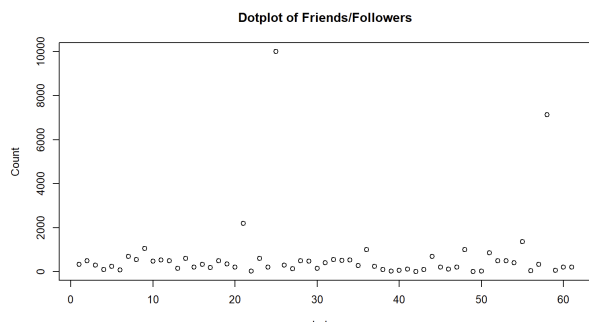
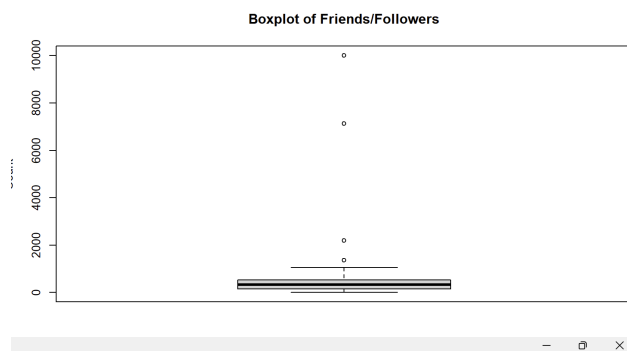
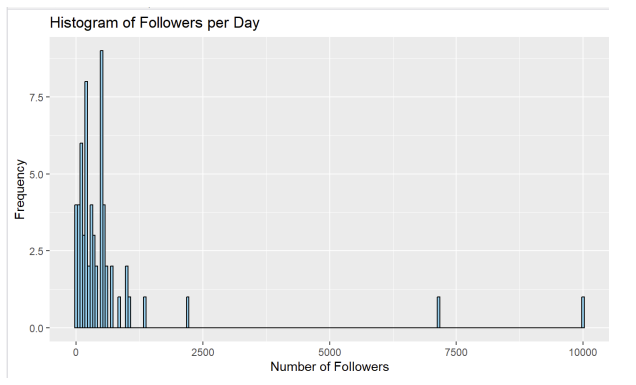
```
Min. 1st Qu. Median Mean 3rd Qu. Max.
10.0  147.0  324.0  660.1  533.0 10000.0
```

Based on the summary data you provided, it seems the maximum number of friends or followers across your social media platforms is 10,000.

To calculate the ratio, you can divide this maximum value by the mean:

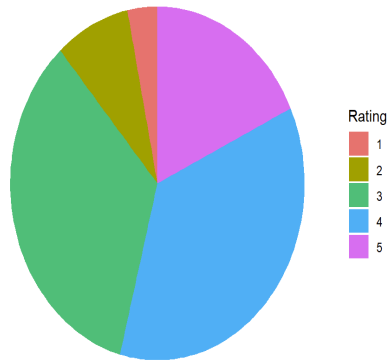
$$\text{Ratio} = \text{Max} / \text{Mean} = 10,000 / 660.1 \approx 15.15$$

So, the highest amount of friends or followers is roughly 15 times the average amount.

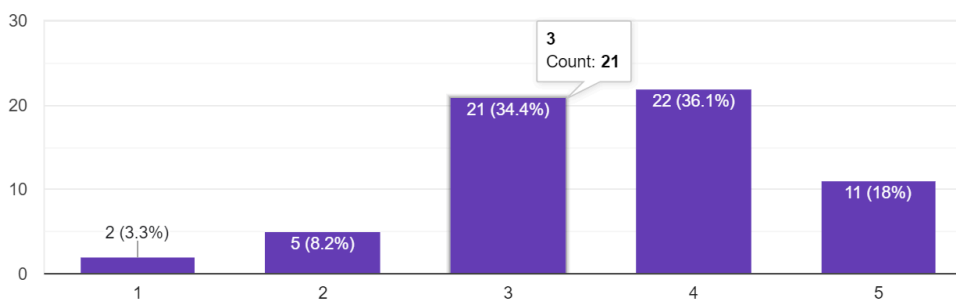
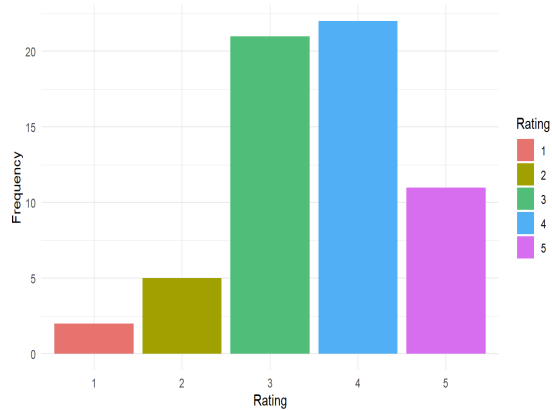


3.12 How would you rank the importance of social media in your daily life?

Importance of Social Media in Daily Life (Pie Chart)

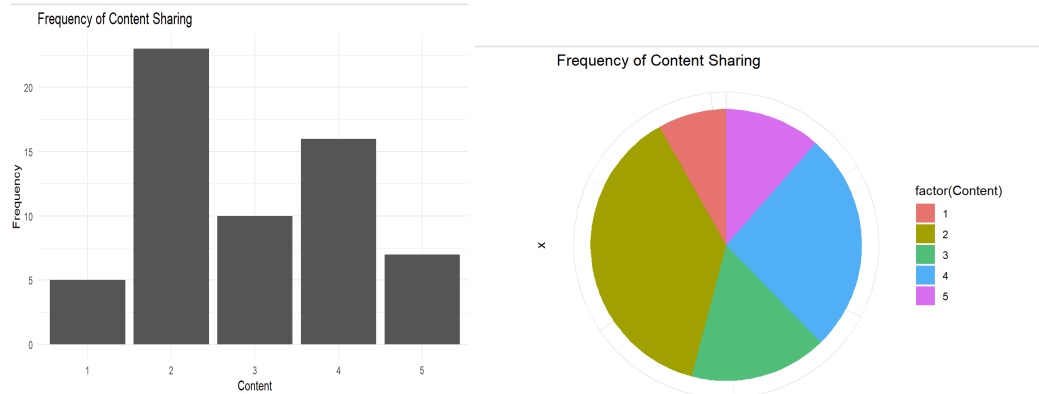


Importance of Social Media in Daily Life (Bar Chart)



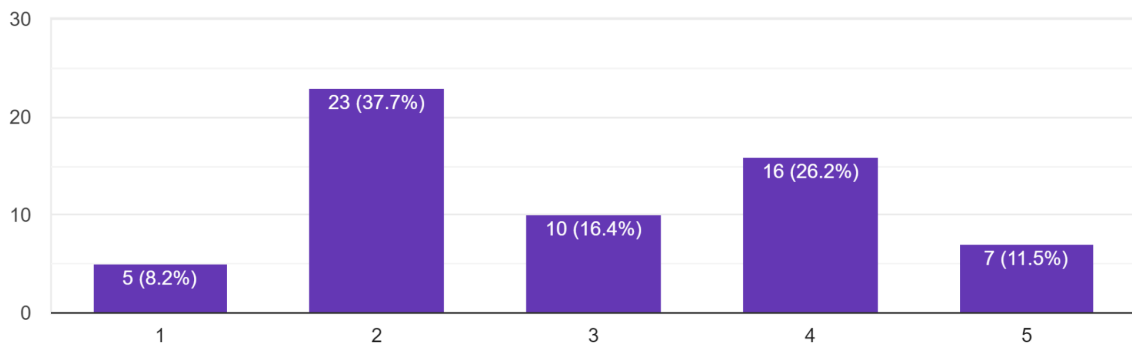
Among the 61 interviewed samples, 11 people agreed that social media is the most important, accounting for about 18%, which is the largest proportion. Followed closely by 22 respondents who believed that social media has important 4 out of 5, accounting for 36.1%. Later, 21 respondents had 3 out of 5, representing approximately 34.4% of the respondent base. Meanwhile, 5 respondents, accounting for 8.2%, strongly agree that 2 out of 5. 2 respondents strongly disagree that social media is important, with the proportions of 3.3%.

3.13 How often do you share the contents on social media? (text post, image, video)



How often do you share the contents on social media? (text post, image, video, etc)

61 responses



Among the 61 interviewed samples, 7 people rate 5 out of 5 to post content in social media, accounting for about 11.5%, which is the largest proportion. Followed closely by 16 respondents who post social media 4 out of 5, accounting for 26.2%. Later, 10 respondents had 3 out of 5, representing approximately 16.4% of the respondent base. Meanwhile, 23 respondents, accounting for 37.7%, strongly post social media that 2 out of 5. 5 respondents rarely post on social media, with the proportions of 8.2%.

4.0 CONCLUSION

The findings of the survey that was carried out among university students about the use of social media show that our survey is more likely to reflect the situation of 20-22 year old students, which is relevant to our survey's situation. When compared to male respondents, there are more female respondents (%) who took part in our questionnaire survey. Additionally, given that the majority of respondents were Chinese, we may claim that this data is well-presented for the use of social media for Chinese students that study in Malaysia's university. The majority of replies come from the UTM, which implies our data cannot accurately reflect any other university other than UTM. All of our responses are university graduates, varying in years and academic backgrounds. We can infer from this data that degree students in Malaysia university are accurately represented.

The result of the study, according to the analysis, shows that 26 out of 61 respondents often use social media for communication and entertainment. It shows that university students are using social media wisely as they mostly use social media for good purposes. In addition to prove this statement, 26 out of 61 respondents choose Telegram as the most used social media in their daily life. From his result we can infer that Telegram is a good platform for communication and entertainment as the respondents can contact people by using Telegram. Besides, they can relax by playing games on Telegram.

According to the data analysis, 37.7% of respondents rarely share contents on social media whatever in the form of text post, image or video. It demonstrates that most university students don't like to share anything related to their daily life on social media. To prove this statement, according to the data analysis, 45.9% of respondents do not post anything on social media. This suggests a prioritization of personal privacy among university students, possibly influenced by network size or relationship status. More than 50% respondents show that the highest number of friends or followers among their social media platforms is less than 500 people. From this data, we can say that our respondents are quite small and this may cause them to seldom share content on social media.

The result of the study also shows that more than 50% of respondents spent less than 5 hours on social media platforms per day. From this result, we can conclude that university students are not prone to become addicted to social media. In addition, most of our respondents think that social media plays an important role in their daily life. From rank 1 to 5, our respondents mostly vote 4 for the importance of social media in their daily life.

In conclusion, social media plays a vital role in our daily life. We use It for many purposes such as for education, entertainment, communication and so on. For the majority of university students, social media is very useful in their daily life. Therefore, it is advised that university students should have high self-discipline in playing social media so that they won't be too obsessed with social media.